Slide 1: Title – Bayesian Volatility at Scale

- Today I will be presenting ‘Bayesian Volatility at Scale’.
- The project asks one question from the paper by Magris and Iosifidis (2023): can Variational Inference give Bayesian GARCH estimates and forecasts that are as good as MCMC and quasi-maximum likelihood, and what does it cost to compute them?
- To answer it, I re-ran the paper’s Microsoft case study end to end in Python, compared four estimators on the same data, and then repeated the exercise across the Dow Jones stocks.

Slide 2: MCMC as the Gold Standard

- Bayesian GARCH estimation has traditionally relied on Markov Chain Monte Carlo.
- On the left, MCMC explores the posterior with a random walk. It is exact in the limit, but the draws are sequential, the proposal has to be tuned, and the chains need diagnostics. In high-dimensional models it mixes slowly.
- On the right, Variational Inference turns estimation into an optimisation problem: we maximise the ELBO with momentum or Adam, monitor a single convergence curve, and the same machinery scales to high-dimensional machine-learning models.
- One thing to say up front: as we will see, for a small three-parameter GARCH, MCMC is not actually slow. VI’s advantages lie elsewhere.

Slide 3: Recasting the Problem as Optimisation

- First, the true posterior $p(\theta \mid y)$ is intractable because the marginal likelihood $p(y)$ has no closed form for GARCH.
- Second, we pick a tractable family $q(\theta)$, here a Gaussian with mean μ and covariance Σ.
- Third, we make q as close as possible to the true posterior by minimising the Kullback–Leibler divergence. That replaces sampling with optimisation.

Slide 4: The ELBO Objective

- Minimising the KL divergence is equivalent to maximising the Evidence Lower Bound, the ELBO.
- The first term, the expected log-likelihood, rewards fitting the data. We estimate it by Monte Carlo: 50 draws per step for BBVI and 10 for the full-covariance method.
- The second term, the KL divergence to the prior, keeps q close to the prior and acts as the regulariser. This is important later: the prior is already in the objective, so we do not need to add an extra penalty on top.

Slide 5: Parameter Transforms

- A Gaussian lives on the whole real line, but GARCH parameters are constrained: $\omega > 0$, α and $\beta \geq 0$, and $\alpha + \beta < 1$ for stationarity.
- Following Proposition 2 of the paper, we optimise an unconstrained vector θ and map it to the GARCH parameters: $\omega = \exp(\theta\omega)$, $\alpha = \text{sigmoid}(\theta\alpha)$, and $\beta = \text{sigmoid}(\theta\beta) \times (1 - \alpha)$.
- Because β is a fraction of the room left by α, $\alpha + \beta$ is below one for every θ. Stationarity holds by construction.
- This replaces my first draft, which rescaled α and β with an if-statement whenever their sum passed 0.99. That rule was not smooth and was not the paper’s method; the transform is smooth everywhere, so gradients are always well defined.

Slide 6: The VI Training Loop

- The loop has five steps: sample θ from q, compute the GARCH likelihood for each draw, add the prior term, compute the gradient, and update the variational parameters.
- I implemented two versions. BBVI is the paper’s Algorithm 1: it uses the score-function, or log-derivative, trick, so it only needs likelihood values and never the likelihood’s derivatives. It uses a diagonal covariance and the paper’s settings: 50 draws, learning rate 0.005, momentum 0.4, and 2,500 iterations.

- The second is full-covariance Gaussian VI in PyTorch. It uses the reparameterisation trick, $\theta = \mu + L\varepsilon$, so that autodiff can differentiate straight through the GARCH recursion, with 10 draws and Adam.
- Both target exactly the same posterior with the same $N(0, I)$ prior on θ and the same transforms.

Slide 7: Data Pipeline

- To make the results checkable, I used exactly the paper's case-study data: Microsoft daily prices from 1 January 2018 to 30 June 2023. That is 1,383 trading days, the same number the paper reports.
- The cleaning checks found no missing values, duplicates or non-positive prices; the longest gap between trading days is four days, which is a holiday weekend.
- Returns are 100 times the log price difference, and the split is chronological: 75% for training up to February 2022 and 25% for testing.
- The test period covers the 2022 rate-hike sell-off, so it is more volatile than most of the training period, a genuine out-of-sample test.

Slide 8: Exploratory Analysis

- The exploratory analysis shows the stylised facts that motivate GARCH.
- Returns have fat tails: excess kurtosis of about 9.5, and Jarque-Bera rejects normality.
- Squared returns are strongly autocorrelated: the Ljung-Box statistic is about 890 and the ARCH-LM test rejects no-ARCH. So volatility clusters and is predictable.
- The mean return is small relative to volatility, so, as in the paper, we set the mean to zero.
- Together this points to a GARCH(1,1) for clustering and Student-t innovations for the fat tails.

Slide 9: Replicating the Paper

- This table compares my estimates (left, in teal) with the paper's Table II (right, in gold).
- QML gives $\omega = 0.160$, $\alpha = 0.186$ and $\beta = 0.769$, against 0.15, 0.19 and 0.77 in the paper, and it also agrees with the standard arch package.
- MCMC, BBVI and the full-covariance VI are all within about 0.02 of the paper's numbers, including the posterior standard deviations.
- Persistence $\alpha + \beta$ is about 0.955, which means a volatility shock has a half-life of roughly 15 trading days. The out-of-sample NLL is 764.6 against 761.4 in the paper, so the implementation is validated.

Slide 10: Posterior Quality

- These are the posterior densities of the three parameters. Grey is MCMC, which we treat as the truth; the dashed teal line is full-covariance VI; and gold is diagonal BBVI.
- Full-covariance VI sits right on top of MCMC.
- Diagonal BBVI has the right centre but is far too narrow: its standard deviation for ω is about 55% too small and for β about 35% too small. A diagonal Gaussian cannot represent the correlation between the parameters, so it is over-confident.
- The ELBO confirms this: full-covariance VI reaches a higher bound, which means a closer approximation. The paper reports the same ordering.

Slide 11: Volatility with Credible Bands

- This is MSFT's annualised conditional volatility, with the 90% credible band from full-covariance VI. Everything after the dashed line is a one-step-ahead forecast.
- The model captures the late-2018 sell-off, the COVID-19 spike, which peaks above 150% annualised, and the 2022 regime out of sample.
- The band is the Bayesian benefit. QML gives a single line, whereas here we get a distribution for tomorrow's volatility. The VI band is the same width as MCMC's, while diagonal BBVI's band is about 10% narrower, which is again over-confident.

Slide 12: Accuracy and Cost

- On the left: out of sample, every method is within 1% of QML on NLL, RMSE, MAD and QLIK. This is the paper's main finding: VI loses essentially nothing in forecasting accuracy.
- On the right is the computational cost, and here I want to be honest. For a three-parameter GARCH, the tuned MCMC takes about 0.3 seconds, BBVI about 1 second, and the PyTorch version about 16 seconds, mostly Python overhead in the loop. So VI is not faster here, and the paper does not claim that it is.
- VI's real advantages are that you monitor one ELBO curve, the same hyperparameters worked for every stock without tuning, you get an analytical posterior you can reuse, and it scales to high-dimensional models where random-walk MCMC struggles.

Slide 13: Student-t Extension

- For the extension, I replaced the normal innovations with a standardised Student-t, with $\nu = 2 + \exp(\theta_\nu)$ so that the variance exists.
- The left Q-Q plot shows Gaussian-GARCH residuals are still fat-tailed; with the Student-t, the points line up.
- The posterior mean of ν is about 7.4, with a 90% interval of roughly 5 to 10, which is far from normal.
- The Student-t model wins on the ELBO by about 11 log-evidence units and on test log-likelihood by about 12. Interestingly, the volatility path is almost identical: the gain is in the shape of the forecast distribution, which matters for tail risk and Value-at-Risk.
- Note that my first draft also added a 'MAP penalty' on top of the ELBO. That counted the prior twice and made the losses incomparable, so I removed it. This comparison is like-for-like.

Slide 14: Scaling Up

- Finally, to mirror the paper's 488-stock study on a smaller scale, I ran QML, MCMC and BBVI on the Dow 30 with identical settings.
- Two stocks were dropped in cleaning: Dow Inc. only listed in 2019, and Walgreens no longer has a price history on Yahoo after going private.
- Across the remaining 28 stocks, MCMC and BBVI are on average about 0.1% from QML on test NLL and about 0.3% on QLIK, the same sub-1% picture as the paper's Table I, and QML has the lowest training NLL for every stock, as it should.

Slide 15: Summary and Takeaways

- First, replication: my QML, MCMC and VI estimates match the paper's Table II within about 0.02.
- Second, accuracy and uncertainty: VI forecasts are within 1% of QML, and full-covariance VI reproduces the MCMC posterior, while diagonal VI is over-confident.
- Third, fat tails and cost: Student-t innovations clearly improve the model, and for a small GARCH, MCMC is cheap; VI pays off through generic tuning and scale.
- So Variational Inference is an accurate, well-calibrated alternative to MCMC for GARCH, as long as the variational family is rich enough. Natural next steps are GJR, EGARCH and FIGARCH, the paper's natural-gradient optimisers, and realised-volatility proxies from intraday data. Thank you.

Question 1: Why Student-t, and what happened to the MAP regularisation?

Question: "Why did you move from a Gaussian baseline to Student-t innovations, and why was the MAP penalty removed?"

Answer:

- Even after GARCH lets the variance move, the standardised residuals of the Gaussian model are still fat-tailed, as the Q-Q plot shows.
- The Student-t adds one parameter, $\nu \approx 7$, and improves both the ELBO and the out-of-sample log-likelihood by more than 10 units.
- The MAP penalty in my first draft added $0.1 \times \Sigma\mu^2$ on top of the KL term, so the prior was counted twice and the 'lower loss' was not an ELBO. The KL term already regularises, so I removed the penalty and compared the models on valid criteria.

Question 2: If VI is not faster here, why use it?

Question: “Your MCMC was faster than VI. Doesn’t that undermine the case for VI?”

Answer:

- For three parameters, no method is slow, and a random-walk sampler with a Hessian-tuned proposal is very efficient.
- My MCMC needed a mode search and a Hessian to tune, while VI used identical hyperparameters for all 28 stocks with no tuning.
- The case for VI is scalability and convenience: one convergence curve, an analytical posterior, and good behaviour in high dimensions, for example in multivariate GARCH, where random-walk MCMC mixes poorly. The key accuracy result, parity with MCMC, still holds.

Question 3: Why is diagonal BBVI over-confident?

Question: “Why did the diagonal BBVI underestimate the posterior uncertainty?”

Answer:

- GARCH parameters are strongly correlated in the posterior: more reaction to shocks requires less persistence.
- A diagonal Gaussian cannot represent that correlation, and minimising $KL(q\|p)$ then shrinks q towards the narrow directions of the posterior.
- The full-covariance version fixes this and matches MCMC, which is exactly what the paper shows with its MGVB and EMGVB optimisers.